

PROTOCOL: Library Preparation

v 1.01

This protocol is for the Flongle by Oxford Nanopore Technologies. For other sequencers and flow cells, contact the ONTOLOGY development team (dev@ontology.bio) for advice.

🌀 Bullet 🌀 points show tips for executing the protocol.

(1) Pool and purify PCRs



It is critical to make fresh 70% EtOH every time because evaporation will decrease the concentration of EtOH in a stock solution.



While not necessary, we recommend performing all mag-bead purifications in Eppendorf LoBind 1.5 mL microfuge tubes, as purification will be slightly more efficient compared to regular 1.5 mL microfuge tubes.

- a. Using a multichannel pipette, pool each PCR plate into a single 1.5 mL microcentrifuge tube. Mix well by pipetting or vortexing and label all tubes carefully with both plate number and ONTOLOGY™ batch.
- b. Combine equal volumes of each plate pool such that the final volume will be at least **200 µL** (e.g., for 10 plates take ≥ 20 µL from each). Mix well and take **200 µL** of the single pool into a new 1.5 mL microcentrifuge tube (exact amount is not critical, 200 µL provides an excess of final DNA in most cases). Store excess pools at -20°C in case repeat runs are needed.
- c. Perform magnetic bead cleanup:
 - i. Add 0.8× volume of Ampure XP beads (e.g. for **200 µL** pooled PCR product, add **160 µL** beads). Mix well by pipetting or vortexing.
 - ii. Incubate for **10 minutes** at **room temperature**, mixing occasionally by gently flicking the tube.
 - iii. During this incubation, prepare 70% EtOH by mixing **1.4 mL** 100% EtOH + **0.6 mL** molecular-grade water for 2 mL total, sufficient for this wash and subsequent washes.
 - iv. Place the tube on a magnetic rack and wait **1–2 minutes** until supernatant is clear, then remove and discard the supernatant.
 - v. Leaving tube on magnetic rack, add **500 µL** of freshly prepared 70% EtOH.
 - vi. After a few seconds, remove and discard EtOH.
 - vii. Repeat (v) and (vi) once, for a total of two EtOH washes.
 - viii. After the second EtOH removal, centrifuge tube for few seconds to collect all remaining EtOH and beads at the bottom of the tube. Return the tube to the magnetic rack and, after the beads pellet (~30 s), use a P20 pipette to remove residual EtOH (if present).



- ix. Leave the tube open, on the magnetic rack, at room temperature for 1-2 minutes to allow residual EtOH to dry.
 - i. The pellet will initially appear shiny/glossy when wet.
 - ii. The beads are sufficiently dry when the pellet takes on a matte (non-glossy) appearance but still has a smooth surface.
 - iii. If the pellet starts to show cracking, it is over-dried. Avoid over-drying but proceed immediately if it happens.
- x. To elute DNA, remove the tube from the magnetic rack and add **40 µL** of molecular-grade water. Resuspend beads by pipetting up and down or by flicking the tube.
- xi. Incubate for **10 minutes** at **room temperature**, mixing occasionally by gently flicking the tube.
- xii. Return tube to magnetic rack. Wait **2 minutes** until supernatant is clear, then transfer **38 µL** supernatant to a fresh 1.5 mL microcentrifuge tube. Discard the tube with the pellet.

(2) Quantification and End Prep



ONTOLOGY™ has a built-in calculator for step **2b**

- a. Quantify **1 µL** of the eluted DNA using Qubit HS kit following manufacturer's instructions.
- b. Set up End Prep reaction:
 - i. From the Qubit result in **2a**, calculate the volume of DNA needed for ~100 ng (this is a guide, not a critical amount). Transfer this volume to a fresh 0.2 mL PCR tube. Unused purified pool may be stored at -20°C for future use.
 - ii. Add molecular grade water to make up a total volume of DNA + water to **12.5 µL**.
 - iii. Combine the following components in the 0.2 mL tube:

	Component	Volume (µL)
1	DNA amplicons (~100 ng)	12.5
2	NEBNext Ultra II End Prep Buffer	1.75
3	NEBNext Ultra II End Prep Enzyme	0.75
	TOTAL VOLUME	15

- iv. Carry out the following program in a thermocycler:

20°C for 5 minutes, 65°C for 5 minutes, hold at 4°C.
- c. Purify the End-Prep reaction:
 - i. Transfer all **15 µL** End Prep reaction to a new 1.5 mL microcentrifuge tube and mix with **15 µL** Ampure XP beads. Mix well by pipetting or vortexing.



- ii. Incubate 5 minutes at room temperature, mixing occasionally by gently flicking the tube.
- iii. Place tube on magnetic rack. Wait 1–2 minutes until supernatant is clear, then remove and discard supernatant.
- iv. Leaving tube on magnetic rack, wash with **200 µL** of prepared-today 70% EtOH
- v. Remove and discard EtOH.
- vi. Repeat steps iv and v once, for a total of two washes.
- vii. After the second EtOH removal, place tube in centrifuge and spin for a few seconds to collect all remaining EtOH at the bottom of the tube. Return tube to magnetic rack and use a P20 pipette to remove any residual EtOH.
- viii. Leave tube open at room temperature for 1–2 minutes to allow residual EtOH to dry (beads are sufficiently dry when they take on a dull appearance; do not over-dry).
- ix. To elute DNA, remove tube from magnetic rack and add **16 µL** of molecular-grade water. Resuspend beads by pipetting up and down or by flicking tube.
- x. Incubate 5 minutes at room temperature, mixing occasionally by gently flicking the tube.
- xi. Return tube to magnetic rack. Wait 2 minutes until supernatant is clear, then transfer 15 µL supernatant to fresh 1.5 mL microcentrifuge tube.

(3) Adapter Ligation and Final Cleanup



ONTOLOGY™ has a built-in calculator for step **4b(xv)**

- a. Combine the following reagents in a 1.5 mL microcentrifuge tube.

	Component	Volume (µL)
1	End-prepped DNA amplicons (~100 ng)	15
2	Ligation Buffer (LNB)*	6.25
3	NEBNext Quick T4 Ligase	2.5
4	Ligation Adapter (LA)*	1.25
	TOTAL VOLUME	25

*From ONT Ligation Sequencing Kit (SQK-LSK114).

- b. Mix by pipetting and incubate at **25°C** for **10 minutes**.
- c. Cleanup final library using Ampure XP beads:
 - i. Add **12.5 µL** (0.5×) volume of Ampure XP beads and mix well by pipetting.
 - ii. Incubate at **room temperature** for **10 minutes**.
 - iii. Place tube on magnetic rack. Wait **~5 minutes** until supernatant is clear, then remove and discard supernatant.
 - iv. Remove tube from rack and add **100 µL** Short Fragment Buffer (SFB, from Ligation Sequencing Kit SQK-LSK114). Pipette up and down or flick tube to resuspend beads.



- v. Return tube to magnetic rack. Wait **3-4 minutes** until supernatant is clear, then remove and discard supernatant.
- vi. Repeat steps **iv** and **v** for a total of two washes in SFB.
- vii. After the second wash, place tube in mini centrifuge and spin for a few seconds to collect all remaining SFB at the bottom of the tube. Return tube to magnetic rack and use a P20 pipette to remove the last SFB.
- viii. Air dry pellet for **no more than 30 seconds**, then add **11 µL** Elution Buffer (EB - from Ligation Sequencing Kit SQK-LSK114) and resuspend pellet by pipetting up and down.
- ix. Incubate for **10 minutes** at **room temperature** or optionally, for increased recovery, place on thermocycler block set to **37°C**.
- x. Return tube to magnetic rack. Wait **2 minutes** until supernatant is clear, then transfer **10 µL** supernatant to fresh 1.5 mL microcentrifuge tube.
- d. Quantify **1 µL** of library using Qubit HS kit (following manufacturer's instructions)

(4) Flow cell loading

- a. Prepare Flongle and carry out Flow cell check:
 - i. Remove one Flongle Flow Cell from the fridge and insert into MinION, ensuring the Flongle Adapter is in place.

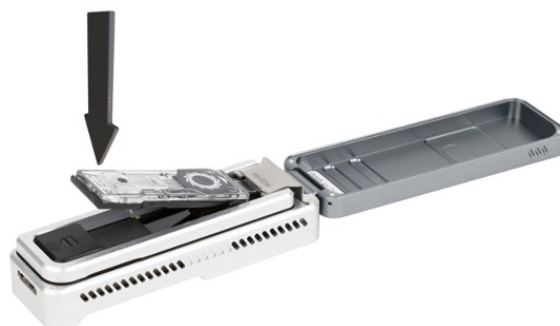


Image credit: Oxford Nanopore Technologies plc.

- ii. In MinKNOW software, navigate to “Flow Cell Check”. Enter the Flongle ID (3 letter+ 3 number code written on the Flongle) and select “FLO-FLG-114” if using 10.4.1 cell.

The check takes ~5 minutes. When complete, record the pore count. If below 50, it is considered to fail warranty and is eligible for a free replacement if within 28 days of receipt; record details and report to ONT.

Note: Pore counts in the range of 20–49 will still often generate high quality results and so you may decide to use a warranty-eligible Flongle rather than wait for a replacement—use your discretion.



- b. While Flow Cell Check is running, prepare flow cell priming mix, prime the flow cell, and prepare the sequencing library mix:
 - xii. In a 1.5 mL microcentrifuge tube, mix **117 μL** Flow Cell Flush (FCF) with **3 μL** Flow Cell Tether (FCT). FCF is from the Flongle Expansion Kit; FCT is from the Ligation Sequencing Kit.
 - xiii. Pull back the plastic film covering the Flongle loading port, holding in place by pressing the dots (adhesive) against the lid of the MinION device.



Image credit: Oxford Nanopore Technologies plc.



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- xiv. Pipette the priming mix (FCF/FT) into the loading port using a P200 pipette. It is important to dispense slowly and not to introduce air bubbles; draw all **120 μL** into the pipette, ensure there is no air gap at the tip and dispense by turning the pipette



plunger rather than pushing down. Leave the last 1-2 μL in the tip and discard to ensure no air is introduced.

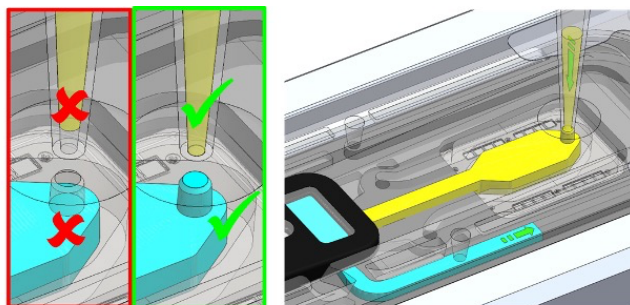


Image credit: Oxford Nanopore Technologies plc.

- xv. From Qubit reading in Step **3d**, calculate volume of library needed for 5 ng DNA in a final volume of 5 μL (ONTOLGY™ contains a calculator for this step). If dilution is necessary, dilute using EB.
- xvi. When ready to load, combine the following components in a 1.5 mL microcentrifuge tube:

	Component	Volume (μL)
1	Sequencing Buffer (SB)	15
2	Library Beads (LIB)*	10
3	DNA library	5
	TOTAL VOLUME	30

**Library beads must be thoroughly mixed by vortexing immediately before use and also mixed by pipetting up and down several times before taking from vial*

- xvii. Pipette library mix up and down several times in the tube to resuspend Library Beads, then pipette into the loading port. NOTE: as above, dispense slowly to avoid introducing air bubbles and leave the last 1-2 μL in the tip.
- xviii. Close the seal tab over the loading port, pressing down on the adhesive spots to stick down. Close the lid of the MinION.

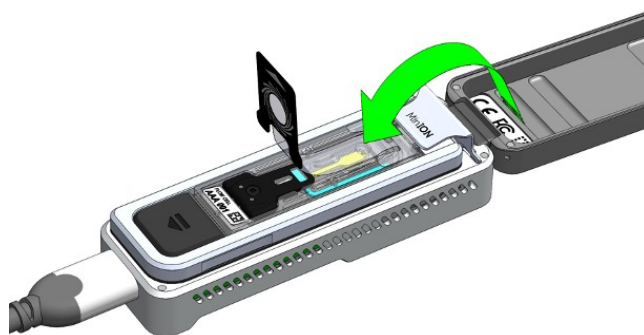


Image credit: Oxford Nanopore Technologies plc.

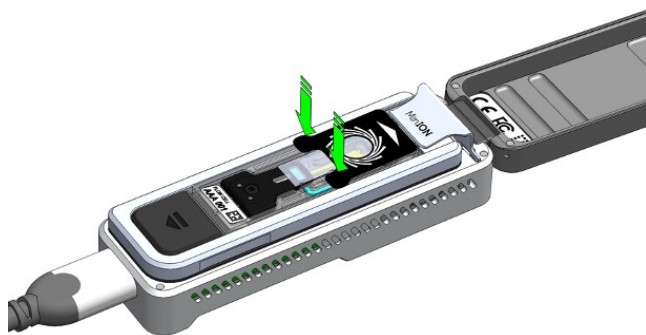


Image credit: Oxford Nanopore Technologies plc.

- xix. Navigate to “Start Sequencing” in MinKNOW software and begin run.

